MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LSFSGU (INCLUDING VARIATION LSFSGA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

Sizes 3 to 1/23 inclusive, extruded silicone rubber insulation

First - Copper conductor, uncoated, class B stranding (see table I for size).
Second - Extruded silicone rubber insulation (see table I for diameter).
Third - Glass braid.
Fourth - Braid covering. Standard identification code applied by method 1.

Sizes 1/23 to 200 inclusive, silicone rubber-glass tape insulation

First - Copper conductor, uncoated, class C and D stranding (see table I for size).
Second - Silicone rubber-glass tape insulation (see table I for diameter). Letter identification code applied by method 5.
Third - The four conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used to form a firm, well-rounded assembly.
Fourth - A binder.

1/ Size 23 may be glass tape or extruded silicone rubber insulation at the manufacturer's option.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LSFSGU

Sizes 3 to 1/ 23 inclusive, extruded silicone rubber insulation

Fifth - The four conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used to form a firm, well-rounded assembly.
Sixth - Binder or combination binder and barrier.
Seventh - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

Sizes 1/ 23 to 200 inclusive, silicone rubber-glass tape insulation

Fifth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

Construction variant for type LSFSGA

(B) Fifth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking not required.
Sixth - Braided metal armor and paint.

1/ Size 23 may be glass tape or extruded silicone rubber insulation at the manufacturer's option.

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TABLE I. Details.

Military part no. M24643/17 1/	Size 2/	Conductor size AWG	Type's overall diameter (inches)				Diameter over outer insulation tape nominal (inch)	Diameter over braid covering (max) (inch)	Diameter over insulation (min) (inch)	Jacket thickness nominal (inch)	Conductor resistance per 1000 feet at 25°C (max) (ohms)	Insulation resistance per 1000 feet (min) (megohms)	Cold bending mandrel diameter maximum (inches)	Accelerated service loading amperes
			LSFSGU		LSFSGA									
			min	max	min	max								
-01UN	3	16(class B)	---	0.447	---	0.497	----	0.130	0.096	0.030	4.3	500	----	----
(B) -02UN	4	14(class B)	---	.513	---	.563	----	.143	.112	.040	2.68	500	----	----
-03UN	9	10(class B)	---	.630	---	.680	----	.187	.154	.040	1.08	500	----	----
-04UN	23	7(class B)	---	.890	---	.940	0.251	.310	.271	.050	0.543	500	----	115
-05UN	50	3(class C)	---	1.050	---	1.100	.334	----	----	.050	.21	200	13	185
-06UN	75	1(class C)	---	1.240	---	1.290	.407	----	----	.050	.134	175	15	250
-07UN	100	0(class D)	---	1.358	---	1.408	.453	----	----	.060	.106	160	16	295
-08UN	150	000 (class D)	---	1.625	---	1.675	.557	----	----	.060	.0674	135	19	400
-09UN	200	0000 (class D)	---	1.820	---	1.870	.634	----	----	.060	.0536	125	21	465

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSFSGU and LSFSGA.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, maximum	(see table I)
Voltage withstand - volts, root mean square, minimum	
Sizes 3 through 9	3000
Sizes 23 and larger	5000
Insulation resistance - megohms/1000 feet, minimum	(see table I)
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
Abrasion resistance - sizes 3 through 23 (extruded silicone rubber) scrapes, minimum	250
(B) Crack resistance - sizes 3 through 23 (extruded silicone rubber)	No damage
Group B:	
Gas flame - 1 hour	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable, sizes 50 and larger - (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - lbf/in^2, minimum	750
Elongation - percent, minimum	125
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (conductor)	
Sizes 3 through 23 (extruded silicone rubber) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Cable filler removability	No failure
Armor (type LSFSGA only) - conformance to material, construction and coverage	No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Requirements
Cold working (minus 20 ± 2°C)	No damage
Gas flame (3 hours)	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
Accelerated service - sizes 23 and larger (see table I for load current)	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
(B) Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSFSGU-3	2500
LSFSGU-4 and 9	2000
LSFSGU-23 and 50	1500
LSFSGU-75 through 150	1000
LSFSGU-200	500

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NOTE: Not For Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-17)

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